SALES PERFORMANCE DASHBOARD

Power BI and Snowflake – Project Documentation

This document summarizes the data-ingestion, Snowflake connection, Power Query preparation and report-building activities completed for the Sales Performance Dashboard.

1. Objective

The objective was to create an interactive sales report in Microsoft Power BI using sales and customer-order data stored in Snowflake. The report is intended to help users review revenue, order volume, customer activity, product-category performance, payment methods and customer ratings.

2. Data Source and Snowflake Preparation

The source dataset was **orders_customers_merged.csv**. It was loaded into Snowflake and made available through the **SALES** table in the **SASI_SALES_DB** database and **SALES_INGESTION** schema.

A CSV file format and an internal stage were configured. The staged file was then loaded into the destination table using **COPY INTO**. A SELECT query was used to inspect the loaded records.

The Snowflake worksheet screenshot records the stage and file-format references, the COPY INTO statement and a sample query against the SALES table.

3. Connecting Power BI to Snowflake

The Snowflake connector was selected in Power BI. The connection form was populated with the Snowflake account server and warehouse, and Import connectivity mode was selected. The Navigator then displayed the SALES table and a preview of its rows.

Import mode was used so that Power BI would load a copy of the source data into the model. The screenshots show the connection configuration, the Import-mode choice and the Navigator preview.

Note: An earlier Power BI Service connection attempt returned an “Invalid credentials” message. The later screenshots show a successful Snowflake table-navigation and import workflow in Power BI Desktop. For publishing to Power BI Service, Snowflake credentials may still need to be configured separately for the semantic model or cloud connection.

4. Power Query Data Preparation

The SALES query was opened in Power Query Editor. The visible transformation removes REVIEW_COMMENT and EMAIL, leaving 13 columns and 197 rows in the displayed query preview. This reduces the fields carried into the report where those personal or free-text fields are not needed for the current visuals.

The preview shows purchase and delivery dates, product category, price, quantity, payment type, customer and order identifiers, review score and other sales/customer attributes. Missing delivery dates are visible in the source and should remain blank rather than being treated as zero-day deliveries.

5. Dashboard Design and Visuals

The report page is titled **Annual Sales Report**. It uses a compact, single-page layout with KPI cards across the top and supporting visuals arranged below.

Visual	Metric / fields represented	Purpose
KPI cards	Total Revenue, Total Orders, Customers, Average Order Value	At-a-glance sales summary.
Line / area chart	Revenue over purchase date	Show revenue variation across 2024.
Donut chart	Average rating by product category	Compare customer ratings by category.
Table	Payment type and sum of order total	Show totals by payment method.
Horizontal bar chart	Revenue by product category	Compare category-level revenue.

Visual	Metric / fields represented	Purpose
Donut chart	Revenue contribution by payment method	Show payment-method revenue share.
Column chart	Customer distribution across quarters	Compare customer distribution by quarter.
City selector	Customer/order city values	Filter report data by city.

The screenshots show KPI values of approximately 123.61K total revenue, 197 total orders, 126 customers and 627.44 average order value. These are the values displayed in the captured report; currency is not explicitly identified in the source screenshots.

6. Interactivity

A city selector is included on the report page. Power BI visual interactions can also allow selecting a category, payment method or time period in one visual to cross-filter or highlight the other visuals. The screenshots document the city selector; additional slicers and interaction behaviour should be tested in the finished report before submission.

7. Observations and Validation Notes

• The captured model contains one SALES table with 197 rows in the Power Query preview.

• The displayed KPI cards report 197 orders and 126 customers.

• The monthly revenue visual is labelled Monthly Revenue, but its axis appears to show many date points across the year. Confirm that the X-axis uses a month-level date hierarchy or a Year-Month field if a monthly summary is intended.

• The chart titled AVG RATING by PRODUCT_CATEGORY displays average rating, not revenue. It should not be described as category revenue contribution.

• The payment-type table and payment-method donut are both useful, but their measure should be clearly labelled as revenue/order total.

• The quarterly customer chart should be checked to determine whether it counts distinct customers or counts order rows. Use DISTINCTCOUNT(CUSTOMER_ID) if the intended metric is unique customers.

• Test the city selector and cross-filter interactions, and confirm that all KPI cards and charts respond as expected.

8. Conclusion

A Snowflake-backed sales dataset was made available to Power BI, prepared in Power Query and used to create an initial Sales Performance report. The report includes summary KPIs, revenue trends, category comparisons, payment analysis, customer distribution and a city selector. The next recommended steps are to validate each measure and interaction, refine the date axis to monthly granularity, add any required category/date/customer slicers, and publish the report with valid Snowflake refresh credentials.

9. Screenshots and Evidence

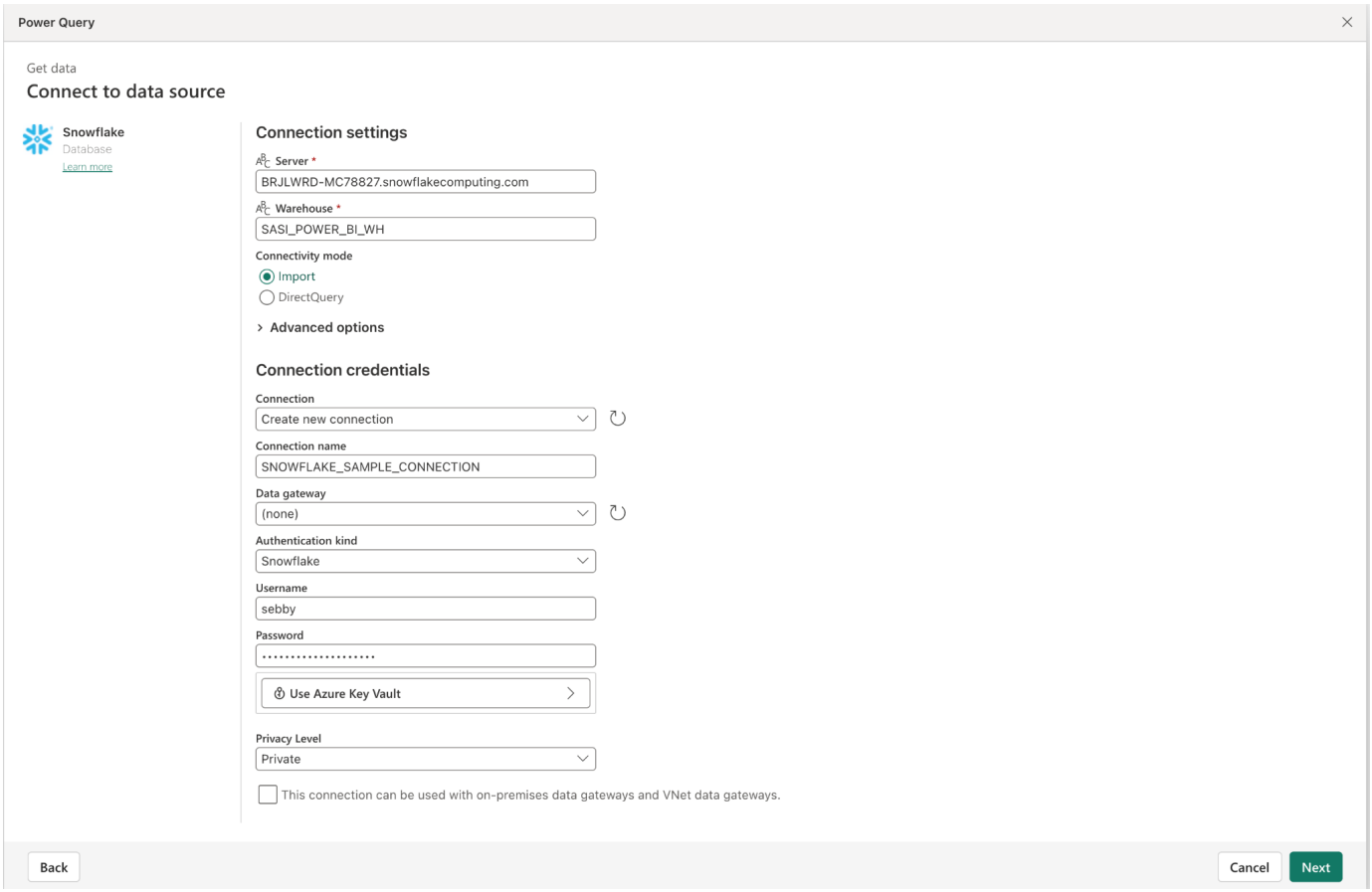


Figure 1. Power Query Online Snowflake connection settings. The server, warehouse, Import mode and Snowflake authentication fields were configured.

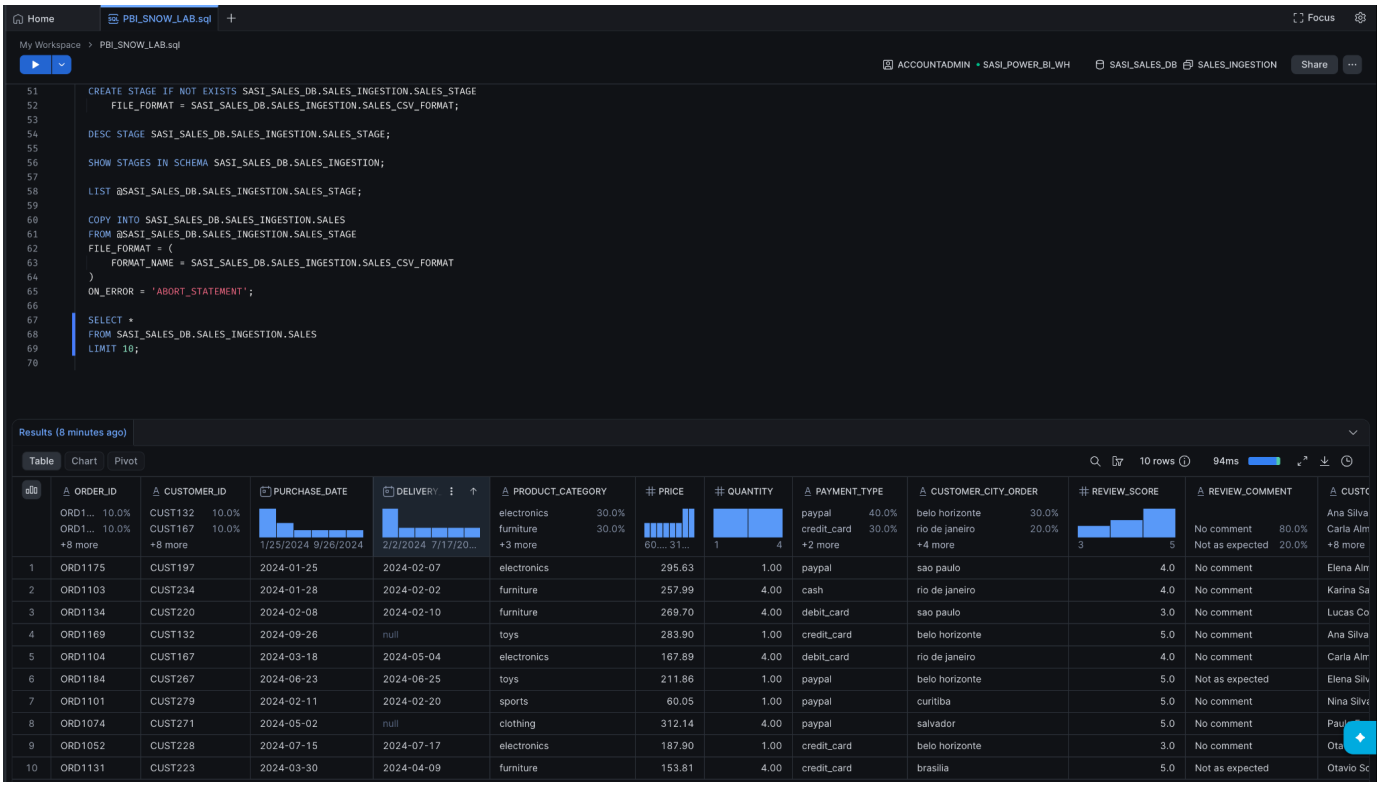


Figure 2. Snowflake worksheet showing the stage, file format, COPY INTO command and a query to inspect the loaded SALES table.

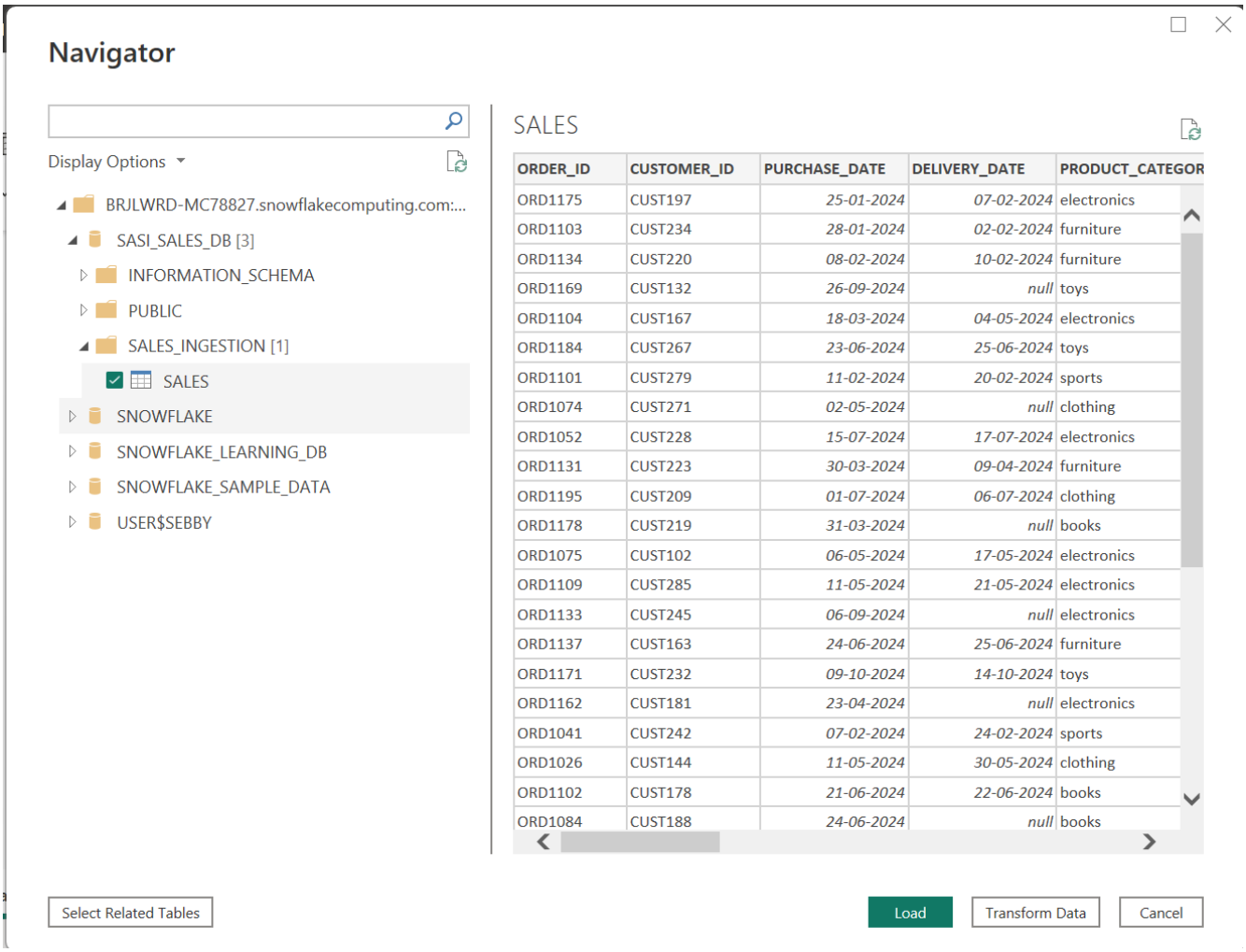


Figure 3. Power BI Navigator displaying the SALES table in the Snowflake database.

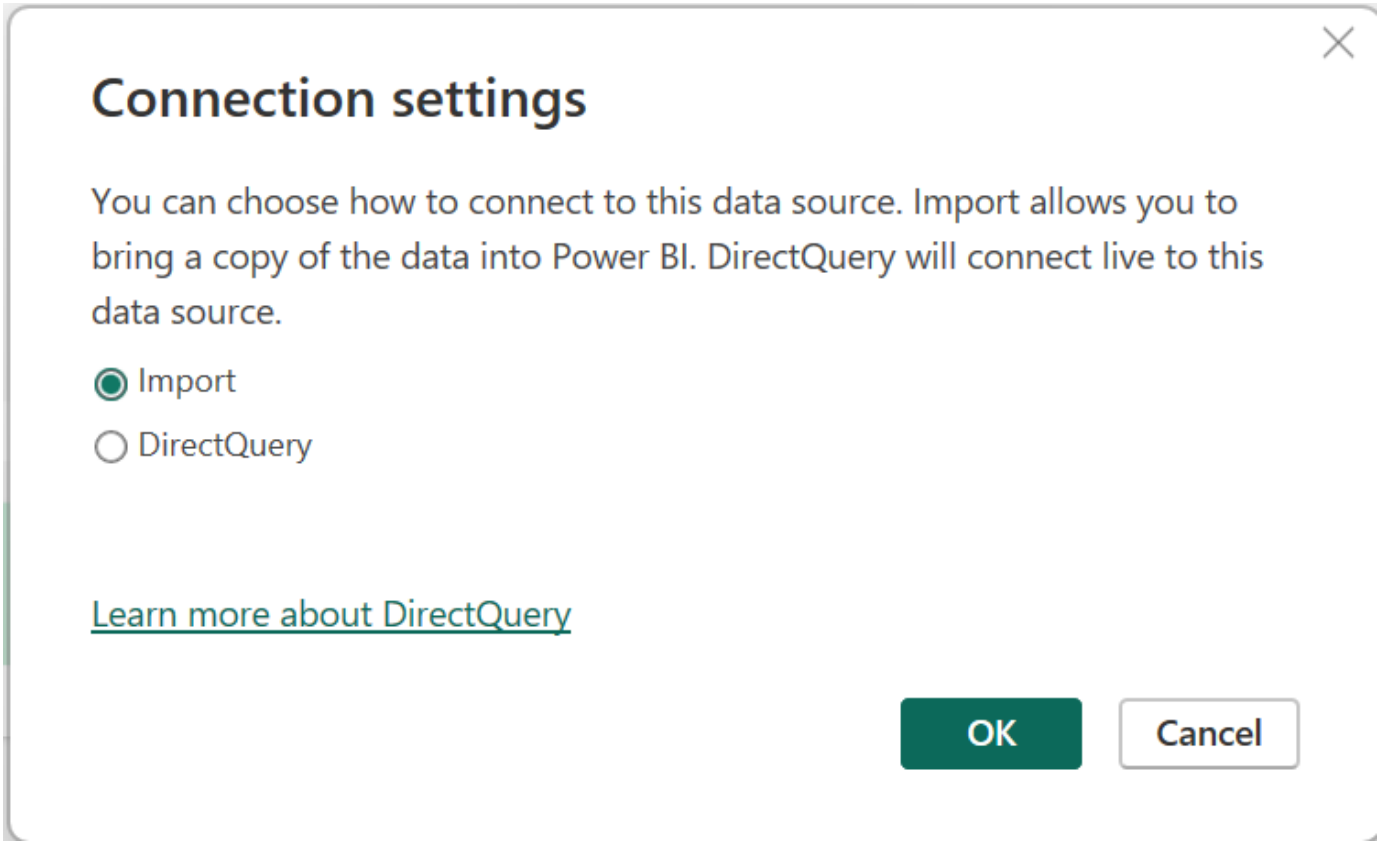


Figure 4. Import mode selected for the Snowflake connection.

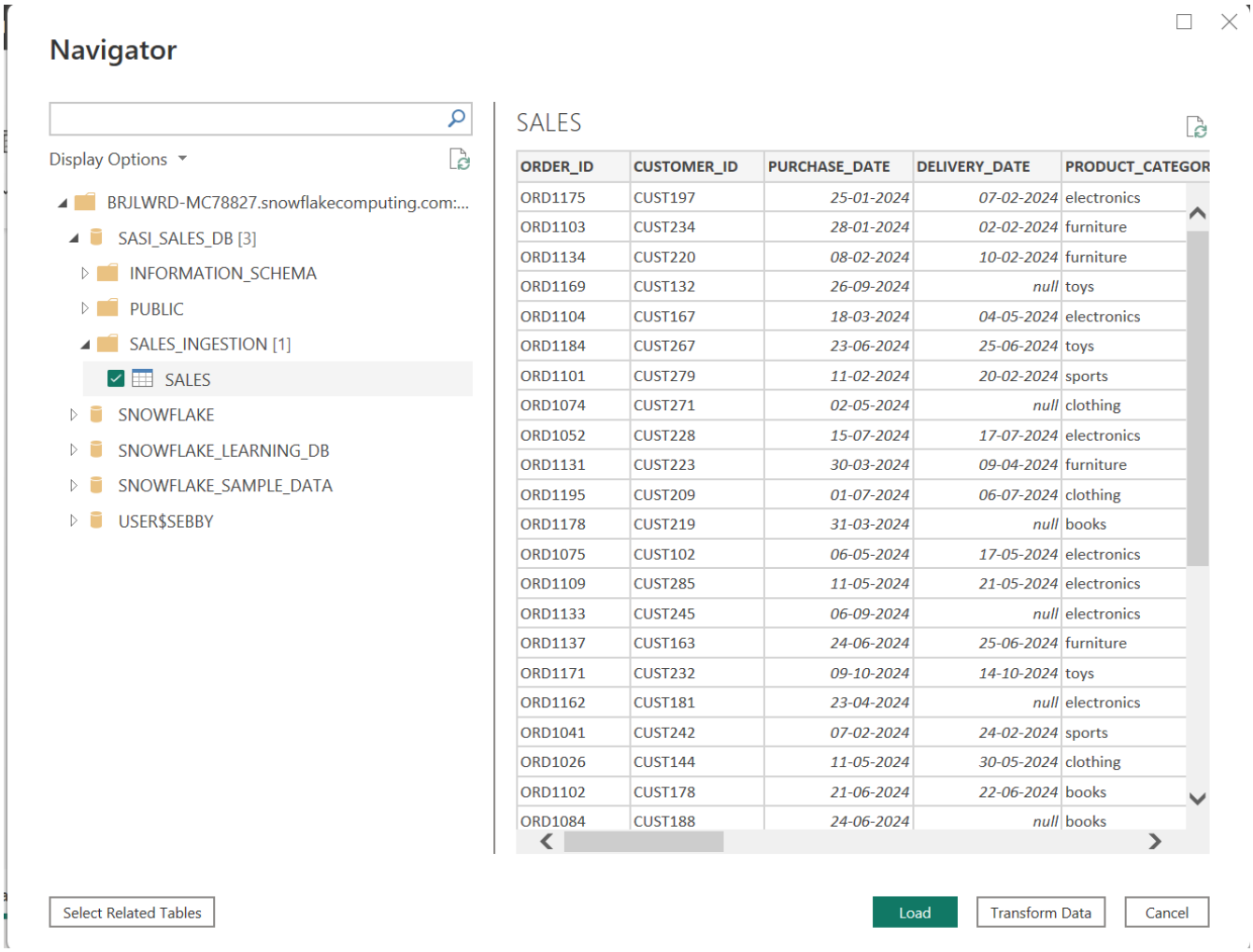


Figure 5. Navigator with the SALES table selected and its data preview visible.

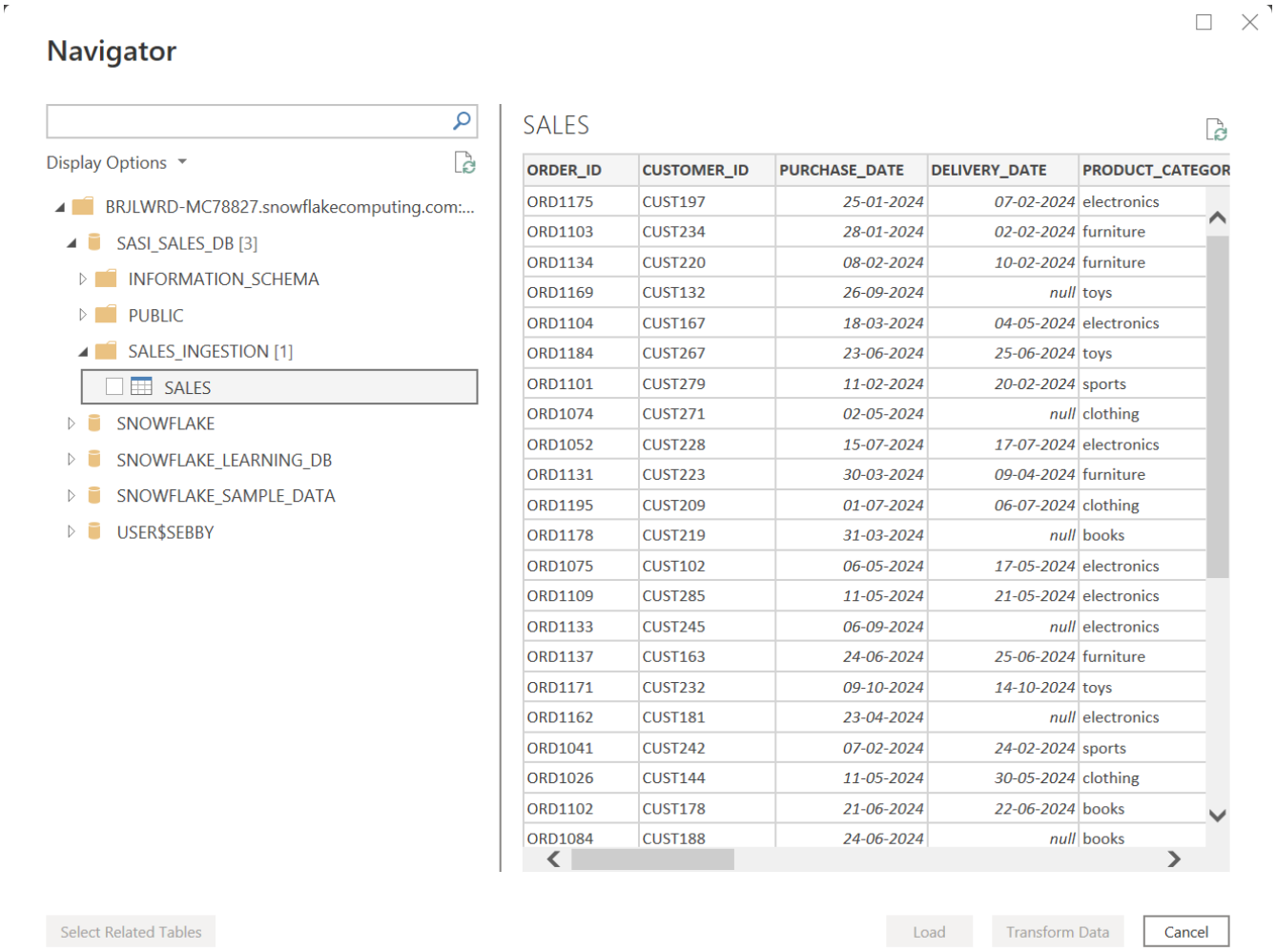


Figure 6. Navigator after the table selection was cleared; this illustrates the table-selection step before loading.

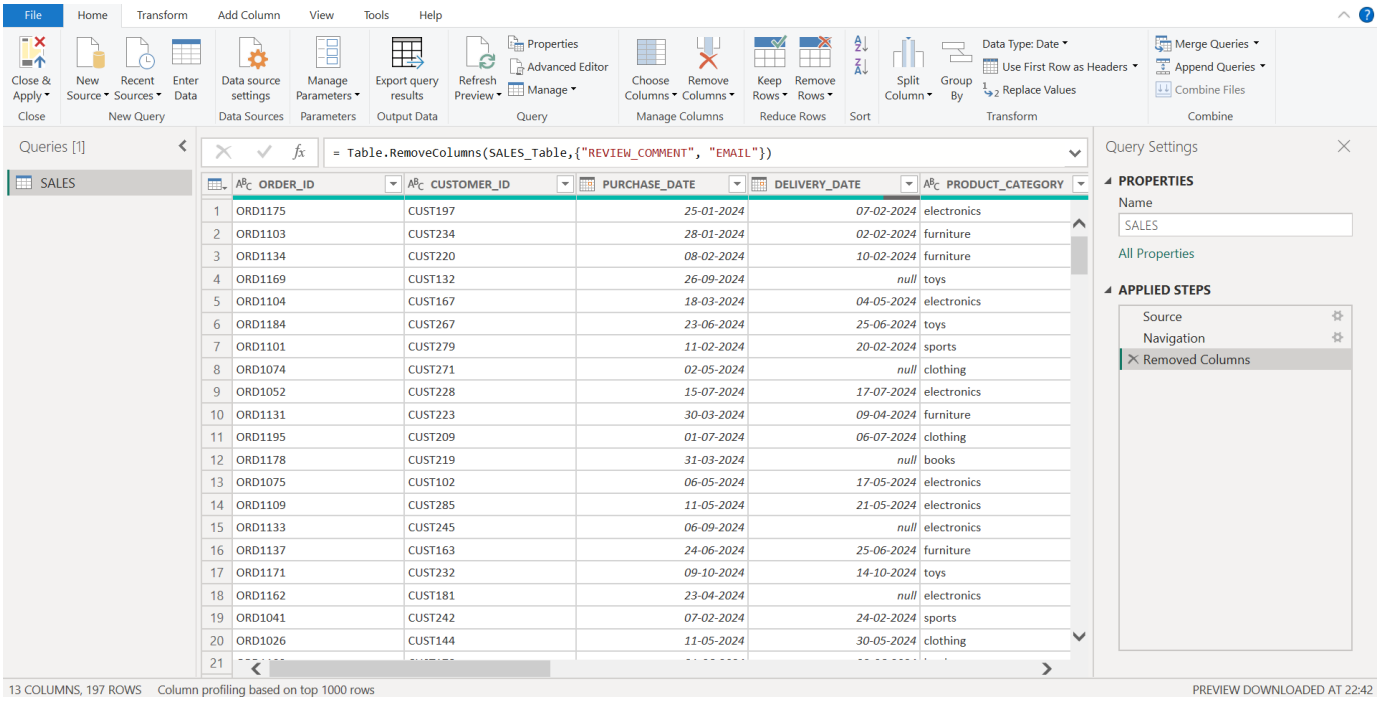


Figure 7. Power Query Editor showing the SALES query and a transformation that removes REVIEW_COMMENT and EMAIL.

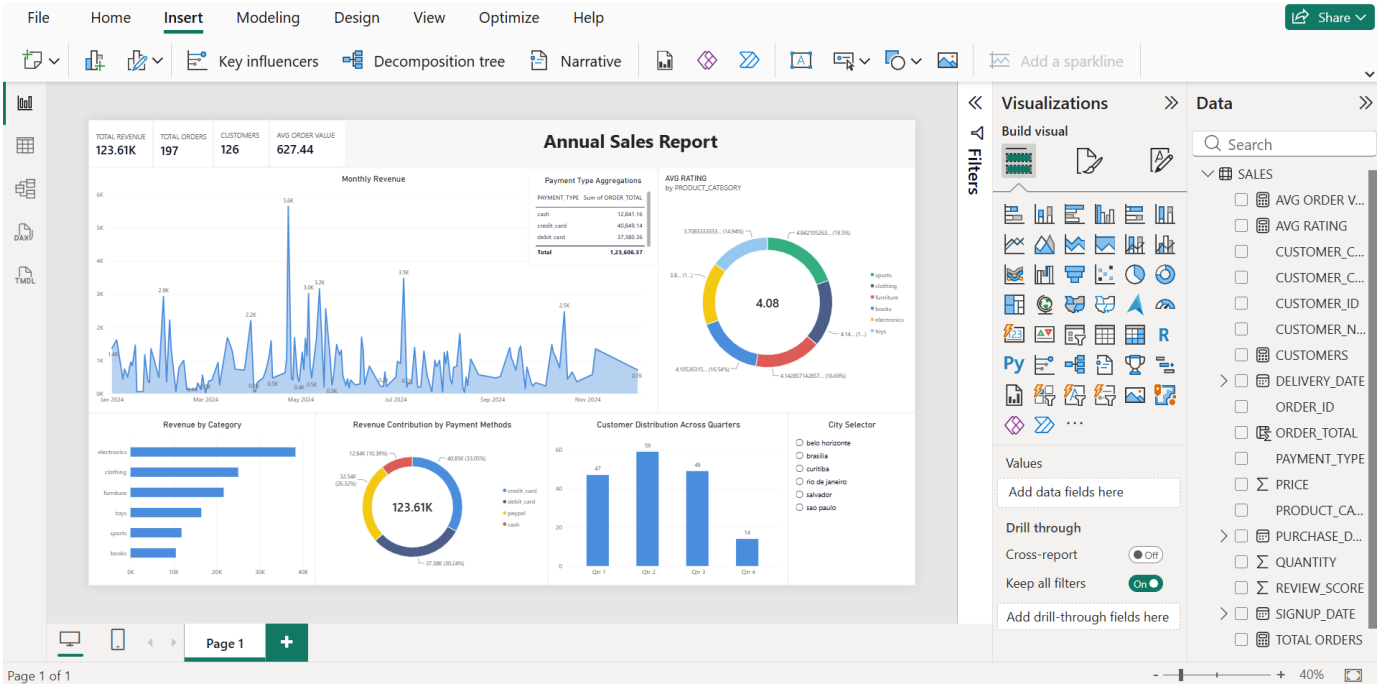


Figure 8. Sales report page showing KPI cards, monthly revenue, rating by category, payment-type aggregation, category revenue, payment contribution and quarterly customer distribution.

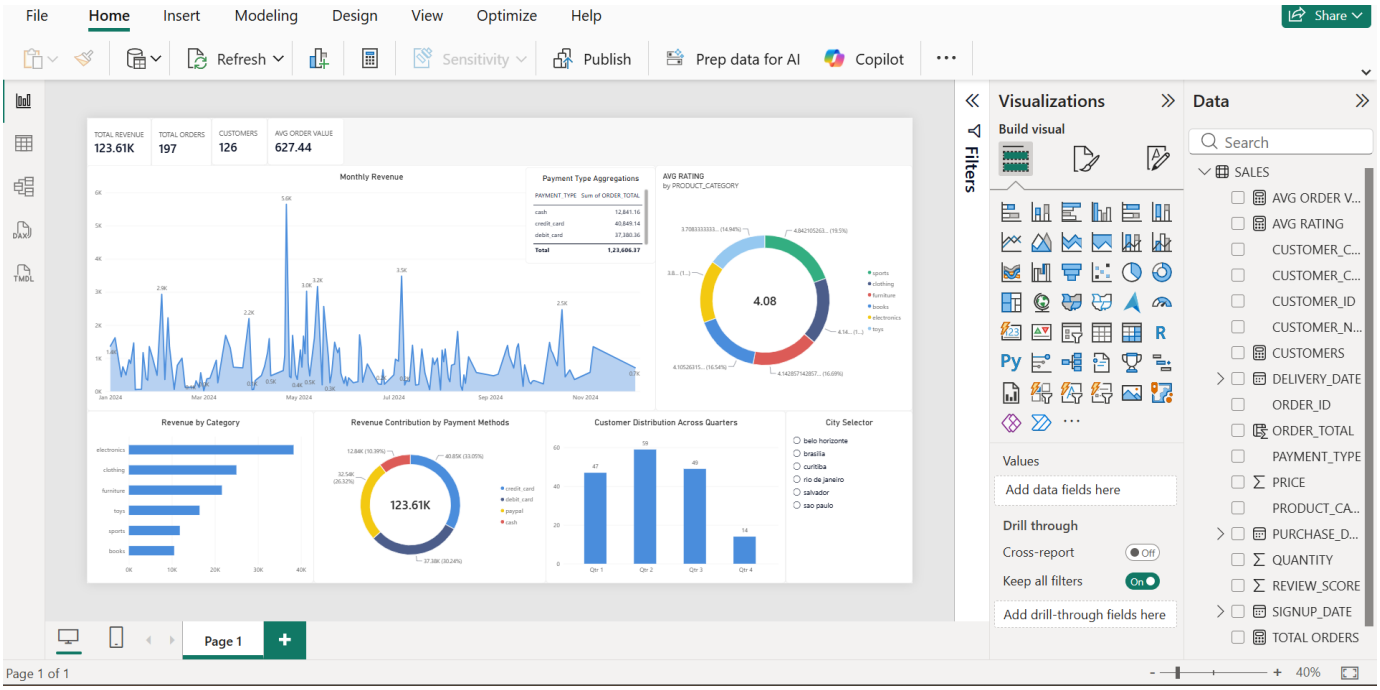


Figure 9. Final report canvas in Power BI Desktop with the same visual layout and the Visualizations and Data panes available for further editing.